

Design of Biocompatible Wearable Antenna Using EBG Structure for Low SAR Application

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Abstract - This paper describes the design and performance analysis of a biocompatible, wearable, microstrip patch antenna of 2.45 GHz operation for application in healthcare as well as wireless systems for body-centric communications. The antenna is designed on a flexible Polydimethylsiloxane (PDMS) substrate, thus making it mechanically flexible and comfortable for the wearer. The problem of electromagnetic field absorption by biological tissues is mitigated through incorporation of an Electromagnetic Band Gap (EBG) structure beneath the antenna structure, which serves to suppress surface waves as well as backward radiation. The use of a three-layer human phantom tissue model comprising skin, fat, and muscle for evaluating antenna performance is recommended. The presented antenna has achieved -25 dB of return loss at 2.45 GHz operation and a notable SAR attenuation relative to other antennas.

Keywords - Wearable antenna, simulation, PDMS substrate, Electromagnetic Band Gap (EBG), Specific Absorption Rate (SAR), healthcare application.

1. INTRODUCTION

Wearable Antennas play a vital role in the development of modern health care technology, wireless body area networks, and smart monitoring devices. When the wearable antenna is placed in contact with the human body, the electromagnetic parameters of the antenna are significantly affected due to the high permittivity and conductivity of the human body. This results in frequency detuning, impedance mismatch, the excitation of surface waves, and the increase of electromagnetic absorption. The rate of absorption of the electromagnetic wave by the human body is quantified by the specific absorption rate (SAR). As per the IEEE standards, the SAR for the average mass of 1 g of the body should be less than or equal to 1.6 W/kg. Due to the development of new materials, flexible substrates such as Polydimethylsiloxane

(PDMS) are being investigated for wearable antenna design due to their low permittivity, flexibility, and biocompatibility. However, the propagation of surface waves towards the body is still a major problem for wearable antenna design. Electromagnetic Band Gap (EBG) structures are promising for the reduction of SAR, whereas artificial magnetic conductors can be employed for SAR reduction and backward radiation reduction. In this work, a wearable antenna design is proposed, which incorporates a compact EBG for SAR reduction at a frequency of 2.45 GHz. [1] Generally, wearable antennas that operate near the human body experience detuning effects, reduced radiation efficiency, and an increased Specific Absorption Rate (SAR). Several researchers have addressed the above-mentioned issues by making the substrate flexible or by employing Electromagnetic Band Gap (EBG) technology. [2] presented a flexible patch antenna for wearable healthcare monitoring systems. the antenna experiences detuning effects, reduced gain, and an increased SAR due to its near-body operation. It also emphasized the need for structural modifications to maintain the stability of the antenna performance. [3] investigated the possibility of using PDMS as a flexible substrate for wearable antenna design. Its concluded that the flexibility of the substrate and the effects of substrate deformation significantly affect the impedance matching of the antenna. PDMS showed promising results for wearable antenna substrate due to its flexibility. [4] presented a wideband wearable antenna with reduced Specific Absorption Rate for biomedical application at a frequency of 2.45 GHz. Optimizing the ground plane and the antenna geometry reduced the Specific Absorption Rate significantly. [5] By integrating the proposed EBG structure into the antenna design, the surface waves were reduced, and the backward radiation to the human body was minimized. Based on the literature survey carried out above, it can be concluded that although the flexibility of the substrate helps to enhance the usability of the antenna, the problem of reducing the SAR level for wearable electronics is a major challenge. Therefore, the proposed work focuses on the design of a compact PDMS-based wearable antenna integrated with an EBG structure to reduce the level of SAR to 2.45 GHz. presented a comprehensive analysis of the effects of the placement of the antenna and the ground plane on the performance of the wearable

antenna operating in the ISM band. a patch antenna design that evaluated the performance of the proposed antenna while the antenna is placed on different body locations, i.e., the arm, chest, with a conventional metallic ground plane and artificial ground-plane structures like EBG. It was noted that antennas with conventional ground planes showed significant backward radiation and near-field coupling with biological tissues. This increased the absorption of electromagnetic radiation by the body, causing an increase in the value of SAR and performance degradation due to impedance detuning effects. On the other hand, the application of the proposed design with the inclusion of artificial ground plane structures effectively reduced the propagation of surface waves and backward radiation towards the body. The suggested artificial ground planes serve as an insulating layer between the antenna and biological tissues. This reduces the amount of electromagnetic field penetration into the human body without compromising the impedance matching, gain, and stability of the radiated field. Therefore, it can be said that the suggested artificial ground planes have an essential role in designing wearable antennas to comply with international SAR guidelines.

Due to the growing trend in wearable healthcare technologies and WBANs, there is now a great need for small-sized and flexible and safe antenna designs. Since wearable antennas are designed in close proximity to the human body[8], the coupling between electromagnetic fields and human tissue becomes highly significant. This phenomenon causes high SAR, poor radiation efficiency, impedance mismatch, and possible biological effects on human beings. The traditional substrate materials such as FR4 cannot be used in wearable antennas because they are non-flexible and lead to poor electrical performance when bent. Hence, it is desirable to use biocompatible and flexible materials like PDMS in wearable antenna design. Yet, despite using flexible materials, the absorption of electromagnetic energy by the human body tissue is still a key problem. The first goal of this research is to create a low-SAR wearable antenna that operates at 2.45 GHz through embedding an EBG structure underneath the antenna. It is expected that this antenna structure will minimize the effect of surface waves and backward radiation to the human body while keeping the SAR level low. The objective of this project is to contribute to the development of better wearable antennas for healthcare applications.

The remaining part of this paper will be presented in the following way: Part 1 will contain the introduction and background of wearable antennas and the necessity of lowering SAR values in body-centric communication applications. Part 2 will discuss the methodology behind the design of a PDMS-based microstrip patch antenna. Section 3 is for presenting the results of the simulations carried out on the proposed antenna design. Section 4 is for presenting the conclusion of the proposed antenna design. Finally, Section 6 is for presenting the references cited in the research work.

2. METHODOLOGY

The design and analysis of the proposed wearable antenna were carried out entirely through simulation. The methodology consists of three major stages: antenna design, integration of the Electromagnetic Band Gap (EBG) structure, and evaluation using a human tissue phantom model.

2.1 Design of the Wearable Patch Antenna

The antenna was designed to operate at the Industrial, Scientific and Medical (ISM) band frequency of 2.45 GHz. A rectangular microstrip patch configuration was selected due to its low profile, ease of fabrication, and suitability for wearable applications.

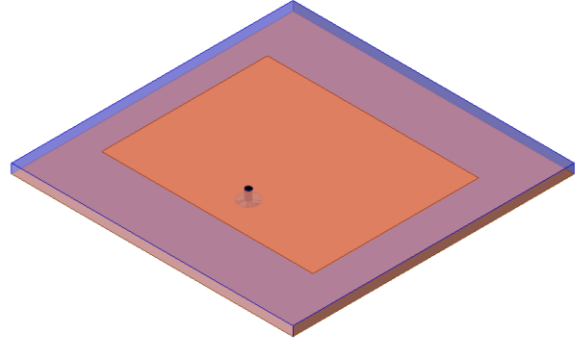


Fig.1 PDMS-based patch antenna operating at 2.45 GHz.

the designed microstrip patch antenna on PDMS substrate. In this, a rectangular patch is kept on the top of the flexible substrate, while the ground plane of the patch antenna is kept at the other side of the substrate. A coaxial probe feed is used to feed the patch antenna at a suitable position to achieve impedance matching at a frequency of 2.45 GHz for ISM band operation. PDMS substrate has been used in this patch antenna because of its flexible properties as well as biocompatibility. The designed patch antenna has been used as a main antenna in the design of the antenna for efficient operation, while a 4x4 EBG has been used to reduce the surface waves as well as SAR while kept in contact with the human body.

Table 1.1: Microstrip patch antenna dimensions

Parameter	Dimension
Ground plane	60X60mm
Substrate	60X60mm
Substrate thickness	2mm
Patch	45X35.3mm

The choice of substrate for the proposed wearable antenna is polydimethylsiloxane (PDMS). PDMS was chosen for this antenna because it has excellent mechanical flexibility and biocompatibility, making it suitable for body-centric or wearable antenna applications. PDMS is a flexible substrate with a low dielectric constant, which is beneficial in maintaining a constant antenna response while enhancing antenna efficiency[9]-[12]. Furthermore, PDMS can be used for a bent condition, making it a suitable substrate for flexible electronic devices. In terms of the simulation for the proposed antenna in ANSYS HFSS software, the electrical properties of PDMS substrate materials were set at a relative permittivity (ϵ_r) of 2.7, a loss tangent ($\tan\delta$) of 0.02, and a substrate thickness of 2 mm.

To start with, the initial dimensions of the rectangular microstrip patch antenna were efficiently computed using the transmission line model equation. Initially, the width of the patch was efficiently computed to ensure efficient radiation at a specific frequency of operation. In addition, the effective dielectric constant was efficiently computed to ensure efficient radiation between the substrate and the air due to the fringing effect of the fields. The effective length of the patch was efficiently computed using the above-mentioned effective dielectric constant. The extension of the patch in terms of length was efficiently estimated, and the final dimensions of the patch were efficiently computed accordingly. After the optimization process, the final dimensions of the patch were efficiently obtained as 44 mm width, 36 mm length, and the ground plane size is 80 * 80 mm. A coaxial probe was efficiently used to feed the patch, and efficient impedance matching was ensured at 50 Ω using the coaxial probe[5],[7]. The position of the coaxial probe was efficiently optimized to ensure efficient radiation at a frequency of 2.45 GHz, corresponding to the return loss. The initial dimensions of the rectangular microstrip patch antenna were determined using the transmission line model equations described by Balanis [6]

2.2 EBG unit cell design

The design of the 3D model of the EBG unit cell, which has been designed using PDMS. The design of the EBG unit cell involves the use of metallic patches,[13-16] which are used for the purpose of blocking the propagation of surface waves and minimizing any backward radiation that may occur toward the human body.

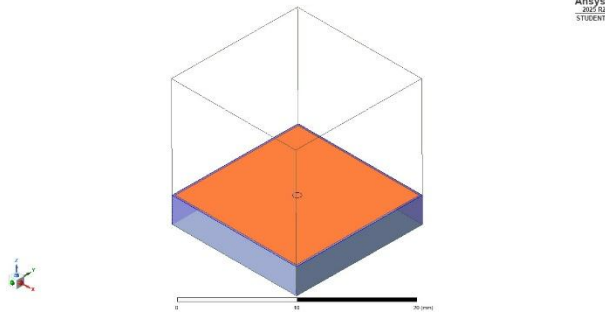


Fig.2.single unit cell

To reduce surface wave propagation and backward radiation, an EBG structure was designed. The EBG unit cell consists of a square metallic patch printed on a PDMS substrate backed by a ground plane. The final design parameters are summarized in Table 2.1

Table 2.1 Design parameters:

Parameter	Dimension
Patch	14.6X14.6mm
PDMS substrate	15X15mm
Ground plane	15X15mm
Via pin	0.4mm
Substrate thickness	3mm

Reflection Phase Analysis

The electromagnetic characteristics of the EBG structure were analyzed using ANSYS HFSS. Periodic boundary conditions were applied with master and slave boundaries on the lateral sides of the unit cell to simulate an infinite periodic structure.

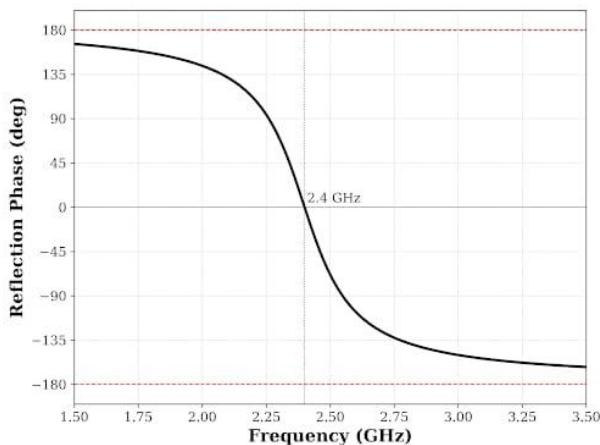


Fig.3.reflection phase

A Floquet port was defined on the top surface to excite the unit cell with a plane wave, and a frequency sweep was performed between 2 GHz and 3 GHz. The reflection phase was plotted to identify the bandgap frequency. The EBG exhibits stopband behavior when the reflection phase crosses 0° , with the bandgap region defined as

$$-90^\circ < \phi < +90^\circ$$

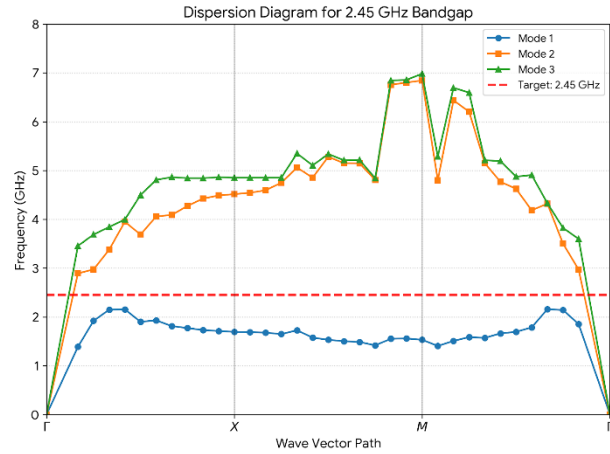


Fig. 4. Dispersion diagram for 2.45Ghz bandgap

The simulated dispersion diagram of the EBG unit cell shows the relationship between the wave vector and frequency along the high-symmetry path. The red dashed line indicates the target operating frequency of 2.45 GHz.

The dispersion curves reveal the existence of a bandgap region where electromagnetic wave propagation is prohibited. This bandgap behavior demonstrates that the EBG structure effectively blocks surface wave propagation at 2.45 GHz, thereby reducing backward radiation toward the human body and lowering SAR values for the wearable antenna.

2.3 INTEGRATION OF 4×4 EBG ARRAY

The 4×4 EBG array was integrated beneath the ground plane of the proposed wearable patch antenna by repeating the unit cell design so that the total EBG surface completely covered the active area of the antenna, thereby ensuring effective suppression of surface wave propagation and backward radiation. This arrangement gave better electromagnetic separation between the antenna and the human body, which is essential in minimizing the SAR value of the wearable antenna system. Throughout the integration process, care was taken to ensure that there was no short circuit formed between the probe feed of the coaxial cable and the EBG structure but still maintained the periodicity of the unit cells .

Following integration, the antenna was re-optimized to achieve impedance matching at the operating frequency of 2.45 GHz. The inclusion of the 4×4 EBG array significantly improved isolation and reduced backward radiation, thereby enhancing both safety and performance.



Fig. 5. structure of mushroom type EBG unit cell

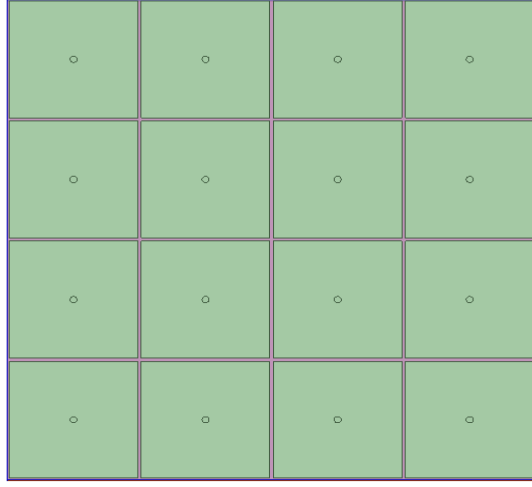


Fig. 6. Layout of 4×4 EBG array used for SAR reduction in the wearable antenna design.

3. HUMAN TISSUES PHANTOM MODELING

To evaluate the on-body performance of the proposed antenna, a three-layer planar phantom was constructed beneath the antenna to represent the electromagnetic properties of the human body. The phantom model consisted of skin, fat, and muscle layers, each assigned frequency-dependent dielectric parameters at 2.45 GHz. The values of relative permittivity (ϵ_r), conductivity (σ), and thickness for each tissue layer are summarized in Table 3.1.

Table 3.1 human phantom model parameter.

Tissue	(ϵ_r)	(σ)	Thickness
Skin	35	1.3S/m	2mm
Fat	5.3	0.1S/m	5mm
Muscle	52.7	1.73S/m	30mm

The antenna was placed directly above the phantom to observe the electromagnetic interaction between the radiating structure and the tissue layers. This setup allowed for the evaluation of Specific Absorption Rate (SAR) distribution and provided insight into the antenna's suitability for wearable applications.

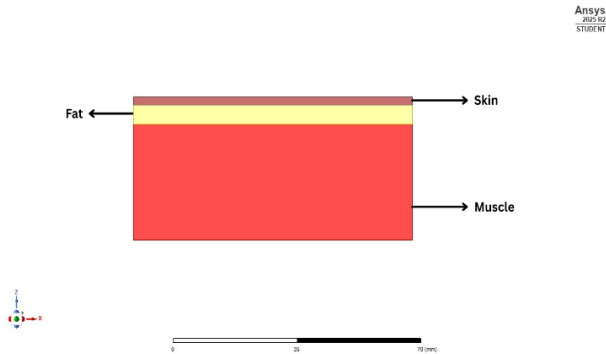


Fig. 7. Human Body Phantom Model

The phantom model ensured that the simulation environment closely replicated realistic human body conditions, thereby validating the effectiveness of the EBG-integrated antenna in reducing SAR levels during on-body operation.

4. RESULTS AND DISCUSSION

4.1 Reflection coefficient

Fig. 8 depicts the comparison of the simulated return loss of the proposed antenna, with and without the 4x4 EBG structure. It can be observed from the above figure that the resonating frequency of the proposed antenna is around the desired 2.45 GHz ISM band and hence ensures good impedance matching.

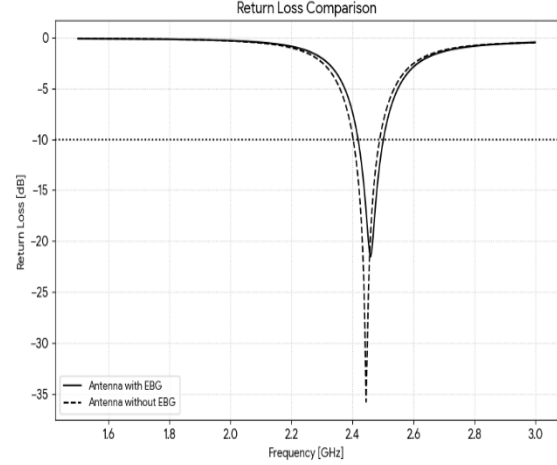


Fig. 8. Simulated reflection coefficient antenna with and without the 4×4 EBG structure.

Moreover, it can be observed that the return loss of the proposed antenna, with and without the EBG structure, is less than -10 dB and hence ensures good antenna performance. It can also be observed from the above figure that there is a slight change in the resonating frequency of the proposed antenna due to coupling between the EBG structure and the antenna. However, it can be observed that there is good impedance matching the proposed antenna at the resonating frequency. This ensures that there is no impact. performance of the antenna due to the integration of the EBG array and hence improves the radiation characteristics of the antenna along with reducing the SAR values.

4.2 radiation characterizes

Fig. 9 illustrates the simulated 2D radiation pattern of the proposed wearable antenna without the 4 × 4 EBG array at the operating frequency of 2.45 GHz.

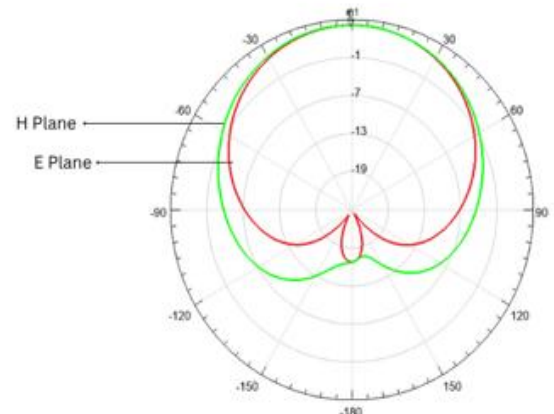


Fig. 9. Simulated 2D radiation pattern of antenna without 4×4 EBG at 2.45 GHz.

It is observed that the antenna exhibits a broadside radiation pattern, with the main beam directed away from the ground plane, which is typical of microstrip patch antennas. The antenna achieves a gain of approximately 4.0 dBi.

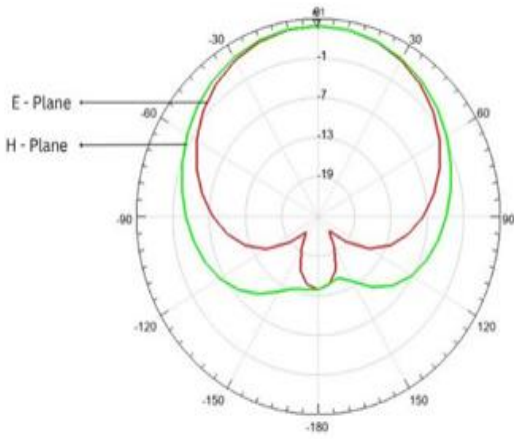


Fig.10. Simulated 2D radiation pattern of wearable antenna with 4×4 EBG at 2.45 GHz.

Fig. 10 shows the simulated 2D radiation pattern of the antenna with the integrated 4 × 4 EBG array at 2.45 GHz. The radiation pattern maintains broadside directionality. However, the inclusion of the EBG array significantly improves the front-to-back ratio. This enhancement reduces backward radiation toward the human body, thereby lowering SAR values while preserving the desired radiation performance. The antenna achieves a gain of approximately 4.1 dBi.

4.3 SAR analysis

The simulated SAR distribution of the wearable antenna without the EBG structure inside the multilayer tissue model is shown in Fig. 11. It can be observed that the maximum SAR value reaches 1.387 W/kg, which is close to the IEEE safety limit of 1.6 W/kg for 1 g averaged tissue.

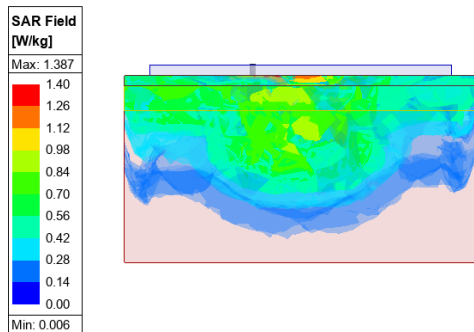


Fig.11. 3D SAR distribution antenna integrated without 4×4 EBG at 2.45 GHz

The SAR distribution is concentrated in the region directly beneath the antenna due to strong electromagnetic coupling, indicating significant absorption of electromagnetic power within the human body.

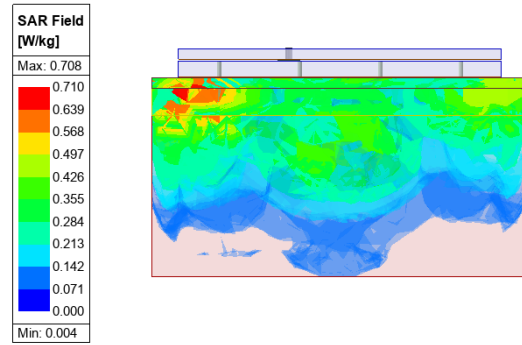


Fig.12. 3D SAR distribution of antenna integrated with 4×4 EBG at 2.45 GHz.

The SAR evaluation was performed using the multilayer phantom model consisting of skin, fat, and muscle layers, with the input power normalized to 1 W. The maximum SAR value obtained in this case is 0.708 W/kg, which is significantly lower than the IEEE safety limit. The SAR distribution demonstrates that most of the electromagnetic energy is confined near the antenna structure, while penetration into deeper tissue layers is reduced.

The implementation of the 4 × 4 EBG structure greatly reduces the effects of the propagation of surface waves, as well as the backward radiation towards the human body. Consequently, the interaction of electromagnetic waves with biological materials is significantly reduced. It has been proven that the proposed wearable antenna design with the use of the EBG structure can operate efficiently while ensuring lower SAR values.

4.4 Comparative study

A comparative study has been conducted the efficiency of the wearable antenna design with the latest research work done on SAR mitigation by employing EBG techniques. [1] wearable antenna integrated with an EBG structure for wireless body area network applications, demonstrating effective suppression of surface waves and improved radiation performance. Similarly, [2] Introduced an EBG-integrated metasurface antenna operating at the 2.4 GHz ISM band, which successfully reduced backward radiation and SAR while maintaining impedance matching.[9] that PDMS is ideal for biomedical applications since it possesses all the necessary characteristics, such as biocompatibility, flexibility, and stability. It also means that PDMS can be used in manufacturing wearable antennas.

Table.4.1 Comparison Table:

Work	Antenna Type	Material Used	Technique Used	Frequency	SAR perform
[1]	Wearable patch antenna	Flexible substrate	EBG and Metasurface	2.4GHZ	0.7W/kg
[2]	Wearable antenna	Conventional materials	Artificial ground plane	~2.4GHz	1.2W/kg
[9]	not antenna (biomedical device)	PDMS	Material fabrication	nil	0.8W/kg
Proposed Work	Microstrip patch wearable antenna	PDMS	4X4 EBG structure	2.45 GHz	0.708W/kg

Uses a flexible PDMS substrate with a 4×4 EBG array structure to obtain improved electromagnetic insulation and SAR reduction [17]. It is demonstrated by simulation that the antenna is operating at 2.45 GHz when impedance matching has been appropriately performed, it shows that the SAR is decreased from 1.387 W/kg without the EBG array structure to 0.708 W/kg due to the application of the 4×4 EBG array, satisfying the IEEE safety level of 1.6 W/kg for 1g averaged tissue.

5. CONCLUSION

This paper proposes a wearable microstrip patch antenna on flexible PDMS substrate operating in the 2.45 GHz ISM frequency band to cater to biomedical and body-centric communication purposes. The conventional patch antenna had relatively more electromagnetic absorption within the human tissue model, leading to increased SAR values due to proximity with the body surface. A 4×4 EBG array was added underneath the patch antenna in order to decrease backward radiation and consequently minimize SAR. The simulation results conducted by ANSYS HFSS proved excellent impedance matching and stable radiation characteristics at the desired frequency. It must be noted that SAR was decreased to 0.708 W/kg compared to 1.387 W/kg without EBG. Consequently, the antenna complies with the IEEE safety limit (1.6 W/kg) for 1 g averaged tissue. Thus, the results obtained prove the effectiveness of introducing an Electromagnetic Band Gap array in order to increase electromagnetic isolation and to minimize SAR. Hence, it can be stated that the proposed wearable microstrip patch antenna proves to be effective in terms of safety and performance. It has great potential for utilization in wearable biomedical and telemedicine devices, wireless body area networks, and body-centred communication systems operating at 2.45 GHz ISM frequency band. However, there is a need to enhance the applicability of the antenna by conducting further research. Future research will aim at proving the efficiency of the proposed antenna via fabrication and measurement. Furthermore, miniaturization methods will be studied in order to optimize the antenna size for better wearability.

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